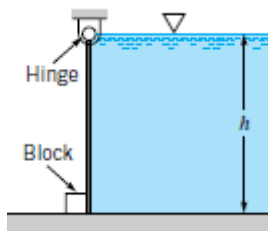


(Use $g = 9.81 \text{ m/s}^2$ and $\rho = 1000 \text{ kg/m}^3$)

- Q1. Two gates of negligible weight are used to hold back water in a channel of width b as shown in FigQ1. The force of the gate against the block for gate (A) is R . Determine (in terms of R) the force against the block for the gate (B).



Gate (A)

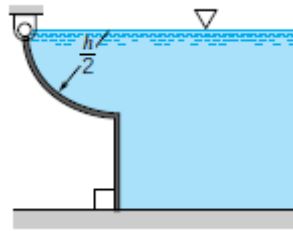


Fig Q1

Gate (B)

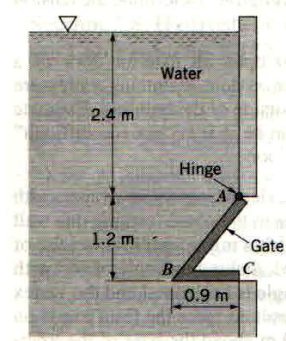


Fig Q2

- Q2. A gate having the cross section shown in FigQ2 closes an opening 1.5 m wide and 1.2 m high in a water reservoir. The gate weighs 2.2 kN and its center of gravity is 0.3 m to the left of AC and 0.6 m above BC. Determine the horizontal reaction that is developed on the gate at C.
- Q3. A converging elbow turns water through an angle of 120° in a vertical plane. The flow cross section diameter is 30 cm at the elbow inlet and 15 cm at the elbow outlet. The axis at inlet is horizontal and the centre of the outlet section is 1.5 m below the centre of the inlet section. Total volume of water in the elbow is 0.9 m^3 . The water volume flow rate is 250 Liters/sec and the elbow inlet pressure is 150 kPa. The elbow mass is 12 kg. Neglecting friction, calculate the magnitude and direction of the anchoring force required to hold the elbow in place.
- Q4. 2.161
- Q5. 3.95
- Q6. 3.161
- Q7. 3.169
- (Problem from Text book 'Frank White', 7th Edition)
- Q8. The device shown in Fig is used to spray an appropriate mixture of water and insecticide. The flow rate from tank A is to be $Q_A = 0.08 \text{ L/min}$ when the water flow rate through the hose is $Q = 4 \text{ L/min}$. Determine the pressure needed at point (1) and the diameter, D , of the device. (Hint: Neglecting kinetic energy of insecticide compared to water at point (1), in SFEE between point 1 and outlet)

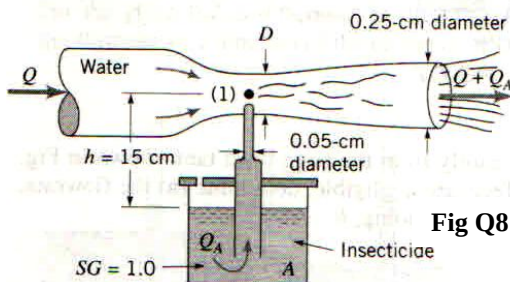


Fig Q8

ANSWERS:

Q1: 1.17R

Q2: 26.82 kN

Q3: Force components on elbow

$R_x = 13.878 \text{ kN} (\rightarrow)$

$R_y = 4.798 \text{ kN} (\downarrow)$

Q8: -24.52 kPa, 0.236 cm